

Prelim 2024: Answer Key

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Question 3 (25 points)

- a) Using the relevant equations above, show that the natural rate of output in this model depends on the markup shock. In particular, show:

$$\hat{y}_t^n = -\frac{1}{\sigma + \psi} \hat{\mu}_t$$

Solution:

The natural rate of output is the level of output one would obtain in a flexible price version of this economy, which requires that

$$\hat{m}c_t + \hat{\mu}_t = 0.$$

Using that $\hat{m}c_t = \hat{w}_t = \sigma \hat{c}_t + \psi \hat{n}_t$, $\hat{c}_t = \hat{y}_t$, and $\hat{n}_t = \hat{y}_t$, we get

$$\sigma \hat{y}_t^n + \psi \hat{y}_t^n + \hat{\mu}_t = 0 \implies \hat{y}_t^n = -\frac{1}{\sigma + \psi} \hat{\mu}_t.$$

- b) Show that the Phillips Curve can be written as:

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \lambda(\sigma + \psi)\hat{x}_t + \lambda\hat{\mu}_t$$

where $\hat{x}_t$ is the welfare relevant output gap, defined as $\hat{y}_t - \hat{y}_t^e$. (**Hint:** in this model the markup shock is the only stochastic element and $\hat{y}_t^e = 0$, which implies $\hat{x}_t = \hat{y}_t = \hat{c}_t$.)

Solution: Using that $\hat{m}c_t = (\sigma + \psi)\hat{y}_t = (\sigma + \psi)\hat{x}_t$, we can rewrite the firms' optimality condition as

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \lambda(\sigma + \psi)\hat{x}_t + \lambda\hat{\mu}_t.$$

- c) Using the method of undetermined coefficients, find the response of the output gap, $\hat{x}_t$, and inflation, $\hat{\pi}_t$, to an exogenous increase in $\hat{\mu}_t$ when prices are sticky and monetary policy follows the Taylor Rule above. To do this, guess that the solution for each variable is a linear function of the shock $\hat{\mu}_t$.

Solution:

First, note that an equilibrium solves the following system of differential equations:

$$\begin{aligned} E_t(\hat{x}_{t+1}) - \hat{x}_t &= \frac{1}{\sigma}(\phi_\pi \hat{\pi}_t - E_t(\hat{\pi}_{t+1})) \\ \hat{\pi}_t &= \beta E_t(\hat{\pi}_{t+1}) + \lambda(\sigma + \psi)\hat{x}_t + \lambda\hat{\mu}_t \end{aligned}$$

Now, let's guess that, in equilibrium,

$$\begin{aligned} \hat{x}_t &= \Lambda_x \hat{\mu}_t \\ \hat{\pi}_t &= \Lambda_\pi \hat{\mu}_t, \end{aligned}$$

for some constants $\Lambda_x, \Lambda_\pi \in \mathbb{R}$. Introducing these expressions into the equilibrium conditions, we get

$$\begin{aligned} \Lambda_x E_t(\hat{\mu}_{t+1}) - \Lambda_x \hat{\mu}_t &= \frac{1}{\sigma}(\phi_\pi \Lambda_\pi \hat{\mu}_t - \Lambda_\pi E_t(\hat{\mu}_{t+1})) \\ \Lambda_\pi \hat{\mu}_t &= \beta \Lambda_\pi E_t(\hat{\mu}_{t+1}) + \lambda(\sigma + \psi)\Lambda_x \hat{\mu}_t + \lambda\hat{\mu}_t \end{aligned}$$

And using that $E_t(\hat{\mu}_{t+1}) = \rho\hat{\mu}_t$, we get

$$\begin{aligned} (\Lambda_x \rho - \Lambda_x) \hat{\mu}_t &= \frac{1}{\sigma}(\phi_\pi \Lambda_\pi - \Lambda_\pi \rho) \hat{\mu}_t \\ \Lambda_\pi \hat{\mu}_t &= [\beta \Lambda_\pi \rho + \lambda(\sigma + \psi)\Lambda_x + \lambda] \hat{\mu}_t \end{aligned}$$

Since these conditions need to hold for any value of $\hat{\mu}_t$, they can be simplified to the following system of linear equations in (Λ_x, Λ_π) :

$$\begin{aligned} \Lambda_x \rho - \Lambda_x &= \frac{1}{\sigma}(\phi_\pi \Lambda_\pi - \Lambda_\pi \rho) \\ \Lambda_\pi &= \beta \Lambda_\pi \rho + \lambda(\sigma + \psi)\Lambda_x + \lambda \end{aligned}$$

so

$$\begin{aligned} \Lambda_x &= -\frac{\frac{1}{\sigma} \frac{\phi_\pi - \rho}{1 - \rho} \lambda}{1 - \beta \rho + \frac{\sigma + \psi}{\sigma} \frac{\phi_\pi - \rho}{1 - \rho} \lambda} < 0 \\ \Lambda_\pi &= \frac{\lambda}{1 - \beta \rho + \lambda \frac{\sigma + \psi}{\sigma} \frac{\phi_\pi - \rho}{1 - \rho}} > 0. \end{aligned}$$

Hence, the guess is verified. A positive markup shock reduces the output gap and increases the inflation rate.

- d) Now suppose that, instead of following the simple Taylor Rule above, we choose $\hat{x}_t$ and $\hat{\pi}_t$ to maximize welfare. In particular, we will derive the optimal policy under **discretion**. Assuming that the steady state is efficient, a planner who cannot commit to future promises solves the following problem period by period:

$$\min_{\hat{\pi}_t, \hat{x}_t} \frac{1}{2} (\hat{\pi}_t^2 + \vartheta \hat{x}_t^2)$$

subject to

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \kappa \hat{x}_t + \lambda \hat{\mu}_t, \quad (1)$$

taking $E_t(\hat{\pi}_{t+1})$ as given, and where $\kappa \equiv \lambda(\sigma + \psi)$. Using the first order conditions of this problem, show

that the planner follows a *targeting rule* of the form

$$\hat{x}_t = -\frac{\kappa}{\vartheta} \hat{\pi}_t. \quad (2)$$

Solution:

Let ζ denote the Lagrange multiplier associated with the constraint. Then, the FOCs are

$$\begin{aligned} (\hat{x}_t) : \quad \vartheta \hat{x}_t &= \zeta \kappa \\ (\hat{\pi}_t) : \quad \hat{\pi}_t &= -\zeta \end{aligned}$$

Combining these two conditions, we get

$$\vartheta \hat{x}_t = -\kappa \hat{\pi}_t \implies \hat{x}_t = -\frac{\kappa}{\vartheta} \hat{\pi}_t.$$

- e) Finally, let's find the optimal policy. Combining equations (1) and (2) (and iterating forward), show that the inflation rate and the output gap are given by

$$\hat{\pi}_t = \frac{\lambda \vartheta}{\kappa^2 + \vartheta (1 - \beta \rho)} \hat{\mu}_t, \quad \hat{x}_t = -\frac{\lambda \kappa}{\kappa^2 + \vartheta (1 - \beta \rho)} \hat{\mu}_t.$$

Is it possible to fully stabilize inflation and the output gap with optimal policy? *Briefly* explain.

Solution:

Combining equations (1) and (2), we get

$$\hat{\pi}_t = \frac{\beta \vartheta}{\kappa^2 + \vartheta} E_t(\hat{\pi}_{t+1}) + \frac{\lambda \vartheta}{\kappa^2 + \vartheta} \hat{\mu}_t.$$

Iterating forward, we get

$$\hat{\pi}_t = \frac{\lambda \vartheta}{\kappa^2 + \vartheta} \sum_{s=0}^{\infty} \left(\frac{\beta \vartheta}{\kappa^2 + \vartheta} \right)^s E_t(\hat{\mu}_{t+s}).$$

Using that $E_t(\hat{\mu}_{t+s}) = \rho^s \hat{\mu}_t$, this expression simplifies to

$$\hat{\pi}_t = \frac{\lambda \vartheta}{\kappa^2 + \vartheta} \frac{1}{1 - \frac{\beta \vartheta \rho}{\kappa^2 + \vartheta}} \hat{\mu}_t$$

or

$$\hat{\pi}_t = \frac{\lambda \vartheta}{\kappa^2 + \vartheta (1 - \beta \rho)} \hat{\mu}_t.$$

Finally, using the relation between $\hat{x}_t$ and $\hat{\pi}_t$ we found in question d), we get

$$\hat{x}_t = -\frac{\lambda \kappa}{\kappa^2 + \vartheta (1 - \beta \rho)} \hat{\mu}_t.$$

Question 4 (25 points)

- a) State the entrepreneurs' problem. Take the first order condition with respect to l_1 . Show that the entrepreneurs will always choose l_1 to maximize profits in period 1, that is, they choose l_1 to solve

$$w_1 = (1 - \alpha)k_1^\alpha l_1^{1-\alpha}$$

Show that this implies that profits are a linear function of the capital stock k_1 , that is,

$$y_1 - w_1 l_1 = R(w_1)k_1$$

for some function $R(\cdot)$ that depends only on w_1 and parameters.

Solution:

The entrepreneurs' problem is

$$\max_{c_0, c_1, k_1 \geq 0, b_1} c_0 + c_1$$

subject to

$$\begin{aligned} c_0 + k_1 &\leq n_0 + b_1 \\ c_1 &\leq k_1^\alpha l_1^{1-\alpha} - w_1 l_1 - b_1 \\ b_1 &\leq \lambda(k_1^\alpha l_1^{1-\alpha} - w_1 l_1) \end{aligned}$$

Let μ_t be the Lagrange multiplier associated to the budget constraint in period t and η the Lagrange multiplier with respect to the borrowing constraint. The first order condition with respect to l_1 is

$$((1 - \alpha)k_1^\alpha l_1^{-\alpha} - w_1)(\mu_1 + \eta) = 0$$

Since $\mu_1 > 0$ and $\eta \geq 0$,

$$w_1 = (1 - \alpha)k_1^\alpha l_1^{1-\alpha}$$

This implies

$$w_1 l_1 = (1 - \alpha)y_1$$

and

$$l_1 = \left(\frac{1 - \alpha}{w_1} \right)^{\frac{1}{\alpha}} k_1$$

Therefore,

$$y_1 - w_1 l_1 = R(w_1)k_1$$

where $R(w_1) = \alpha \left(\frac{1 - \alpha}{w_1} \right)^{\frac{1 - \alpha}{\alpha}}$.

- b) Using the result from a), restate the entrepreneurs' problem as a simpler linear problem, replacing $y_1 - w_1 l_1$ by $R(w_1)k_1$. Characterize the solution to the entrepreneurs' problem with first order conditions.

Solution:

The simplified problem for the entrepreneurs is

$$\max_{c_0, c_1, k_1 \geq 0, b_1} c_0 + c_1$$

subject to

$$c_0 + k_1 \leq n_0 + b_1$$

$$c_1 \leq R(w_1)k_1 - b_1$$

$$b_1 \leq \lambda R(w_1)k_1$$

The FOCs are

$$(c_0) : 1 - \mu_0 \leq 0 \quad (3)$$

$$(c_1) : 1 - \mu_1 \leq 0 \quad (4)$$

$$(k_1) : -\mu_0 + \mu_1 R(w_1) + \eta \lambda R(w_1) \leq 0 \quad (5)$$

$$(b_1) : \mu_0 - \mu_1 - \eta = 0 \quad (6)$$

- c) Show that if $\lambda R(w_1) < 1 \leq R(w_1)$ the entrepreneurs' investment in capital is strictly positive and finite. What happens if $\lambda R(w_1) \geq 1$? What if $R(w_1) < 1$?

Solution:

From (6) we have

$$\mu_0 = \mu_1 + \eta$$

Plugging in (5)

$$\mu_1(R(w_1) - 1) + \eta(\lambda R(w_1) - 1) \leq 0$$

If $R(w_1) < 1$, the FOC with respect to capital is strictly negative, so $k_1 = 0$. If $\lambda R(w_1) \geq 1$, the FOC with respect to capital is strictly positive, so investment is infinite. If $\lambda R(w_1) < 1 \leq R(w_1)$, the FOC is zero and $k_1 > 0$. To see this, note that if $R(w_1)$, $\eta = 0$ and the FOC is equal to zero. If $R(w_1) > 1$, $\eta = \frac{1 - \lambda R(w_1)}{R(w_1) - 1} > 0$, and from (6), $\mu_0 = \mu_1 + \eta > 0$, so $k_1 > 0$.

- d) Show that if the entrepreneurs are constrained, their choice of capital is

$$k_1 = \frac{n_0}{1 - \lambda R(w_1)}$$

Solution:

When the entrepreneurs are constrained, $\eta > 0$, the borrowing constraint binds and therefore

$$b_1 = \lambda R(w_1)k_1$$

From (6) we have

$$\mu_0 = \mu_1 + \eta > 1$$

hence $c_0 = 0$. From the budget constraint

$$k_1 = n_0 + \lambda R(w_1)k_1$$

or

$$k_1 = \frac{n_0}{1 - \lambda R(w_1)}$$

e) Show that the equilibrium level of capital solves

$$k_1 = \frac{n_0}{1 - \lambda \frac{\alpha}{k_1^{1-\alpha}}}$$

Show that k_1 and the entrepreneurs' profits are increasing in n_0 . **Hint:** Remember that we limit attention to the case in which $\lambda R(w_1) < 1 \leq R(w_1)$.

Solution:

From a) we know that

$$R(w_1) = \alpha \left(\frac{1 - \alpha}{w_1} \right)^{\frac{1-\alpha}{\alpha}}$$

$$w_1 = (1 - \alpha)k_1^\alpha l_1^{1-\alpha} = (1 - \alpha)k_1^\alpha$$

where the second equality uses market clearing. Hence, $R(w_1)$ as function of k_1 is given by

$$\alpha \frac{1}{k_1^{1-\alpha}}$$

Therefore, k_1 solves

$$k_1 = \frac{n_0}{1 - \lambda \alpha \frac{1}{k_1^{1-\alpha}}}$$

The LHS is increasing in k_1 , continuous, and goes from zero to infinity. For the RHS we need to use the condition that $\lambda R(w_1) < 1$, which implies that the RHS is well defined only for $k_1 > (\lambda \alpha)^{\frac{1}{1-\alpha}}$. Under this condition, the RHS is decreasing in k_1 , continuous, and goes from infinity to n_0 . Hence, the LHS and RHS intersect and that intersection is unique. Since the RHS is increasing in n_0 , it is immediate to see that k_1 is increasing in n_0 .

f) Assume that the entrepreneurs are constrained. Suppose the government taxes a lump-sum τ to consumers in period 0, and transfers the receipts from the tax to the entrepreneurs. Derive an expression for the utility of consumers and entrepreneurs as a function of τ .

Solution:

The utility of consumers is

$$U_c = e_0 - b_1 - \tau + e_1 + w_1 + b_1 = e_0 + e_1 + w_1 - \tau$$

where e_0 and e_1 are the consumers' endowments.

The utility of the entrepreneurs is

$$U_e = (1 - \lambda)R(w_1)k_1 = (1 - \lambda)\alpha k_1^\alpha$$

where

$$k_1 = \frac{n_0 + \tau}{1 - \lambda \alpha \frac{1}{k_1^{1-\alpha}}} \quad (7)$$

and k_1 is increasing in τ .

g) Show that if n_0 is sufficiently small, a small positive tax increases the utility of both consumers and entrepreneurs.

Solution:

Since k_1 is increasing in τ , and U_e is increasing in τ , that the utility of entrepreneurs is increasing in τ is immediate. On the other hand, for consumers an increase in τ has a negative direct wealth effect and a positive indirect effect through wages. If $\frac{\partial w_1}{\partial \tau} - 1 > 0$, the tax increases the utility of the consumers. But τ affects w_1 only through k_1 . Thus

$$\frac{\partial w_1}{\partial \tau} = \frac{\partial w_1}{\partial k_1} \frac{\partial k_1}{\partial \tau}$$

We have

$$\frac{\partial w_1}{\partial k_1} = \alpha(1 - \alpha)k_1^{\alpha-1}$$

Note that if n_0 is very small, $k_1 \simeq (\lambda\alpha)^{\frac{1}{1-\alpha}}$. Hence

$$\frac{\partial w_1}{\partial k_1} = \frac{1 - \alpha}{\lambda}$$

Moreover, we can use the implicit function theorem to obtain $\frac{\partial k_1}{\partial \tau}$. Rewrite (7) as

$$k_1 - \lambda\alpha k_1^\alpha = n_0 + \tau$$

Differentiating with respect to τ

$$\frac{\partial k_1}{\partial \tau} - \lambda\alpha^2 k_1^{\alpha-1} \frac{\partial k_1}{\partial \tau} = 1$$

Evaluating at $k_1 = (\lambda\alpha)^{\frac{1}{1-\alpha}}$

$$\frac{\partial k_1}{\partial \tau} = \frac{1}{1 - \alpha}$$

Putting everything together

$$\frac{\partial w_1}{\partial \tau} = \frac{1 - \alpha}{\lambda} \frac{1}{1 - \alpha} = \frac{1}{\lambda} > 1$$

So a small tax when n_0 is small increases the utility of both consumers and entrepreneurs.

Question 5 (25 points)

- a) Assume there are no financial markets available, so that individuals must simply invest on their own. Given that an individual has invested an amount I at time $t = 0$, what will be the optimal levels of consumption, c_1, c_2 , if:
- (i) the individual receives a liquidity shock (i.e. is impatient)
 - (ii) the individual does not receive a liquidity shock (i.e. is patient)

Solution: (i) If the agent receives the liquidity shock, she values only consumption in $t = 1$. This implies that she will liquidate all the projects she undertook in $t = 0$:

$$c_1 = \underbrace{1 - I}_{\text{mattress}} + \underbrace{\frac{I}{4}}_{\text{projects liquidated}}$$

(ii) If the agent doesn't receive the liquidity shock, she values consumption only in $t = 2$, so she will not liquidate any project in $t = 1$. Then

$$c_2 = \underbrace{1 - I}_{\text{mattress}} + \underbrace{4I}_{\text{successful projects}}$$

- b) What is the optimal level of investment, I^* ? What is the ex-ante expected utility of an individual?

Solution: Given the consumption decisions we found in a), we can state the agent's problem as

$$\max_{I \geq 0} -\frac{1}{2} \frac{1}{1 - \frac{3}{4}I} - \frac{1}{2} \frac{1}{1 + 3I}$$

The FOC is

$$-\frac{\frac{3}{4}}{(1 - \frac{3}{4}I)^2} + \frac{3}{(1 + 3I)^2} = 0$$

and after some algebra

$$I^* = \frac{2}{9}$$

Consumption is $c_1^* = \frac{5}{6}$ and $c_2^* = \frac{5}{3}$. The ex-ante utility of the agent is

$$U^* = -\frac{9}{10}$$

- c) Now suppose that when types are realized in period 1, this information is publicly observable. Suppose there exists a social planner that individual's entrust all of their endowment to at time 0. The social planner will pay impatient individuals $\hat{c}_1$ in period 1 and patient individuals $\hat{c}_2$ in period 2.
- (i) Solving the social planner's problem, what is $\hat{c}_1$ and $\hat{c}_2$? How much does the social planner invest? That is, what is $\hat{I}$?
 - (ii) What is an individual's ex-ante expected utility now?
 - (iii) Why can the planner improve over autarky? How would your answer change if $L = 1$?

Solution:

(i) The social planner solves

$$\max_{c_1, c_2, I, L \geq 0} \pi u(c_1) + (1 - \pi)u(c_2)$$

subject to

$$\pi c_1 \leq 1 - I + \frac{L}{4}$$

$$(1 - \pi)c_2 \leq 4(I - L)$$

Replacing the constraints (that are satisfied with equality in an optimum) in the objective function, we get

$$\max_{I, L \geq 0} -\frac{1}{4} \frac{1}{1 - I + \frac{L}{4}} - \frac{1}{4} \frac{1}{4(I - L)}$$

The FOCs are

$$\begin{aligned} (I) : & \& -\frac{1}{\left(1 - I + \frac{L}{4}\right)^2} + \frac{1}{4((I - L))^2} = 0 \\ (L) : & \& \frac{1}{4} \frac{1}{\left(1 - I + \frac{L}{4}\right)^2} - \frac{1}{4((I - L))^2} \leq 0 \end{aligned}$$

From the first FOC

$$\frac{4}{(4(I - L))^2} = \frac{1}{\left(1 - I + \frac{L}{4}\right)^2}$$

Introducing this in the second

$$\frac{1}{4} \frac{1}{\left(1 - I + \frac{L}{4}\right)^2} - \frac{1}{\left(1 - I + \frac{L}{4}\right)^2} < 0$$

Thus, $L = 0$. Then, from the first FOC

$$I = \frac{1}{3}$$

It shouldn't be surprising that the planner doesn't liquidate any project: in $t = 0$ she knows exactly how much she will need to distribute in $t = 1$ so it's better if she saves that amount in the mattress rather than liquidating. Moreover, note that the planner invests **more** than the agents when they invest on their own.

(ii) Consumption is $\hat{c}_1 = \frac{4}{3} > c_1$ and $\hat{c}_2 = \frac{8}{3} > c_2$. The agent's ex-ante utility is

$$\hat{U} = -\frac{9}{16} > -\frac{9}{10} = U^*$$

(iii) The planner can improve on autarky for two reasons. First, she can avoid liquidation, which is inefficient. But not only that. If we set $L = 1$, then there is no cost of liquidating, and the planner can still improve on autarky. The reason is that the planner can provide **insurance**. It does so by increasing consumption in $t = 1$ and reducing consumption in $t = 2$.

- d) Now suppose an agent's type is private information, and the social planner can only offer a contract contingent on an individual's announcement of her type at time 1. Furthermore, at time 1, she meets each agent once with the meeting order randomly determined. If individual's report honestly, can the social planner offer the same contract as in question 4.? Is it optimal for an individual to report honestly

when everyone else does? Explain in 1-2 sentences how this planner can be interpreted as a bank.

Solution: Yes. The reason is that if the agents answer truthfully, the allocation $\{\hat{c}_1, \hat{c}_2, \hat{I}\}$ satisfy feasibility in each period. Moreover, it is optimal for agents to answer truthfully. If they are impatient, they don't win anything by saying they are patient since that implies consuming in $t = 2$ rather than $t = 1$. If they are patient and claim to be impatient, they get $\hat{c}_1$, they put it in the mattress, and consume $\hat{c}_1 < \hat{c}_2$, where $\hat{c}_2$ is what they would consume if they wait and every other patient waits.

The planner can be interpreted as a bank that issues demand deposits. The “asking” part of the planner can be mapped to a bank as the revealed preference of withdrawing money from the account: absent a bank run, only impatient agents will do that. Thus, absent a bank run, demand deposits decentralize the planner's allocation.

- e) Suppose all agents fear a bank run, and each agent reports to the bank at time 1 as being impatient.
- (i) How many individuals will get paid by the bank before it runs out of money in period 1? Given this, explain why a bank run can be an equilibrium, i.e., why it is optimal for a “patient” individual to run on the bank when she expects a bank run.
- (ii) Suppose the bank implements a policy of only paying the first π individuals that show up at time 1, and the rest will get paid at time 2 (i.e. it suspends convertibility). Will this eliminate the bank run as an equilibrium? Can you think of any problems with this policy?

Solution: (i) The total amount of resources available to the bank in $t = 1$ is

$$1 - \hat{I} + \frac{\hat{I}}{4} = 3/4.$$

How many agents can get $\hat{c}_1 = \frac{4}{3}$?

$$\# = \frac{\frac{3}{4}}{\frac{4}{3}} = \frac{9}{16} < 1$$

This implies that if a patient agent expects everyone else to run, they should run not to be one of the $\frac{7}{16}$ that won't get anything.

(ii) This policy eliminates the bank run since patient agents know that if they don't run, they will get $\hat{c}_2 > \hat{c}_1$ in $t = 2$ for sure.

The problem with this policy is that it requires to know with certainty π to be able to decentralize the optimal allocation. If π is stochastic, this policy will not deliver first best: if the realization of π is higher than the level used in the bank's policy, then some impatient individuals won't be able to consume at $t = 1$. However, at this level of analysis we don't know if the policy is *second best* or not.